

Proof ownership and blackboard susceptibility

A minimal epistemic toy extension

Model proposal and small experimental check

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Abstract

We add one epistemic coordinate to the sampled-exposure model: the fraction of agents holding sufficient active evidence to solve the task. A two-group model gives an exact finite-round susceptibility and its epistemic slope. We retain the existing availability normalization, illustrate response regimes, and propose an optional 96-call probe. All numerical curves are toy predictions, not fitted experimental results. The scope is susceptibility only.

1 One new mechanism: proof can reduce responsiveness

Fix a designated false target. Let x be its population vote share and $\phi = N_P/N$ the fraction of agents whose active facts suffice to determine the correct answer. Group P holds a proof; group L has incomplete evidence. Validate sufficiency using the frozen task’s symbolic rules, allowing alternative proofs. Use the existing full-proof flag only if its meaning agrees.

An agent in group $g \in \{L, P\}$ has baseline target preference p_{0g} . After controller exposure its preference is

$$p_{1g} = p_{0g} + \delta_g, \quad 0 \leq p_{0g}, p_{1g} \leq 1. \quad (1)$$

For the protective hypothesis, parameterize

$$\delta_L = \delta > 0, \quad \delta_P = r\delta, \quad 0 \leq r \leq 1. \quad (2)$$

Here $r = 0$ means proof holders have no additional response to exposure, $r = 1$ means proof gives no protection, and intermediate r means partial protection. Neither proof ownership nor $r = 0$ requires every proof holder to vote correctly. Estimate the two responses freely first: a negative or larger proof-holder response challenges this protective family.

Closure assumptions.

1. Proof-group membership and response parameters are fixed for one round. Within-round acquisition, reactivation and forgetting are neglected.
2. There are M updates, each selecting one agent uniformly with replacement; normally $M = N$. Both groups share revision probability γ , unchanged by controller exposure.
3. The board has fixed eligible counts B peer posts and, under activation, b controller posts. Exposure is independent of group and vote; at least one controller message is the sufficient exposure summary.
4. Unexposed revision follows the same law under activation and silence. Task, prompts and controller content are fixed for the comparison.

These assumptions isolate proof-dependent responsiveness. They approximate the evolving blackboard; they do not describe its complete implemented dynamics.

2 Closed-form response and epistemic derivative

For a uniform sample of $q \leq B$ messages without replacement,

$$w_b = 1 - \frac{\binom{B}{q}}{\binom{B+b}{q}} \quad (3)$$

is the probability of at least one controller exposure. For shorter boards use sample size $\min(q, B+b)$, the convention $\binom{B}{m} = 0$ for $m > B$, and zero exposure for an empty sample. Here b is actual posts under activation, not an unfilled allowance.

Conditional on revision, the new target indicator is Bernoulli with effective preference $p_{0g} + U w_b \delta_g$. Let $x_{g,j}$ denote target share within group g after j global update opportunities. Uniform focal selection gives

$$E[x_{g,j+1}] = \left(1 - \frac{\gamma}{N}\right) E[x_{g,j}] + \frac{\gamma}{N} (p_{0g} + U w_b \delta_g). \quad (4)$$

The coefficient is γ/N , not γ/N_g : group g is selected only with probability N_g/N . Subtracting active and silent branches after M updates yields

$$\chi_g = w_b \delta_g \Lambda, \quad \Lambda = 1 - (1 - \gamma/N)^M. \quad (5)$$

Since $x = (1 - \phi)x_L + \phi x_P$, population susceptibility is

$$\boxed{\chi_b(x, \phi) = w_b \Lambda [(1 - \phi)\delta_L + \phi\delta_P] = w_b \Lambda \delta [1 - (1 - r)\phi].} \quad (6)$$

An empty group contributes zero through its population weight. No assumption that $x_L = x_P$ is needed: initial group vote shares cancel under the common-rate revision law. Both branches retain autonomous movement.

The epistemic slope is therefore

$$\boxed{\frac{\partial \chi_b}{\partial \phi} = w_b \Lambda (\delta_P - \delta_L) = -w_b \Lambda \delta (1 - r).} \quad (7)$$

Multiplying this slope by 0.1 gives the change in controller response per ten percentage points of proof ownership, holding the model parameters fixed. In existing observational population states, this is effect heterogeneity, not automatically the causal effect of supplying proofs.

Predictions and falsification. Under protection, susceptibility rises with exposure and falls linearly with proof ownership. At $r = 1$ it is independent of ϕ ; at $r = 0$, a fully proof-owning population has zero additional response. Strong nonlinear dependence on ϕ challenges the constant-parameter closure.

There is no explicit x dependence in this equal-compliance submodel. Strong residual x dependence likewise challenges the closure. It should not be hidden by smoothing. The model does not derive ϕ from persistence ρ : measure proof ownership from inventories and initially compare fixed- ρ conditions. If proof status changes substantially within a round, interpret the formula as a short-window approximation.

Algebra check. Iterating the group-mean recursions with unequal initial group vote shares reproduced Eq. (6) to 4.2×10^{-17} in the illustrative setting on the next page. This verifies the calculation, not empirical adequacy.

3 Illustrative regime maps and availability normalization

Use $N = M = 24$, $q = 3$, $B = 18$, $\gamma = 0.6$, $\delta = 0.3$ and $r = 0.2$. Feasible baseline preferences are $p_{0L} = 0.4$ and $p_{0P} = 0.15$, giving exposed preferences 0.7 and 0.21. These values are illustrative, not estimated from the game.

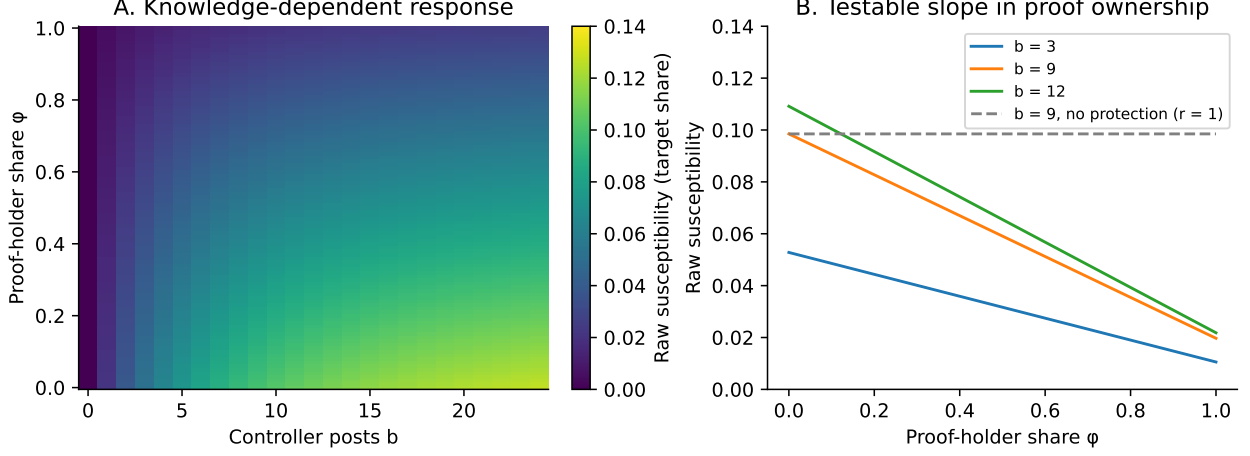


Figure 1. Budget versus proof ownership, colored by raw susceptibility, and representative response curves. The dashed line is a no-protection reference. These are smooth finite-round response regimes, not evidence of a phase transition.

At $b = 9$, exposure is $w_b = 0.7210$. Responses at $\phi = 0, 0.5, 1$ are 0.0985, 0.0591, 0.0197: about 9.85, 5.91 and 1.97 percentage points more false-target support than silence. Proof ownership reduces response while exposure stays unchanged.

Retain the supplied report’s availability-normalized companion:

$$\chi_{b,\text{avail}}(x, \phi) = \frac{\chi_b(x, \phi)}{1 - x}, \quad x < 1. \quad (8)$$

Do not replace $1 - x$ by $1 - \phi$: non-target voters and incomplete-evidence holders are different populations. Mask $x = 1$. This is not a conversion probability: differential retention of existing target voters contributes, so the ratio may exceed one.

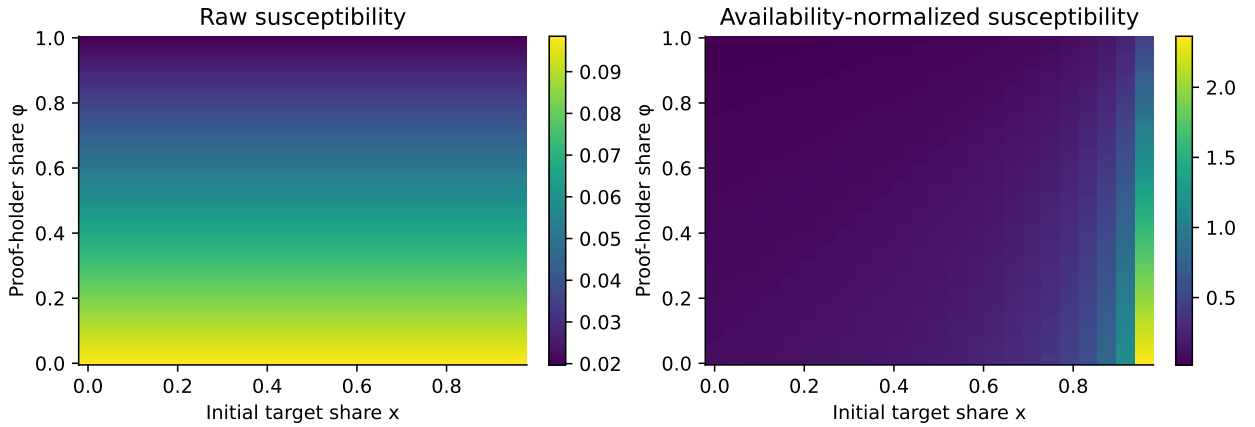


Figure 2. At $b = 9$, raw response is constant across x at fixed ϕ , while availability normalization amplifies high- x values. Color scales differ. This appearance comes from the denominator, not a new control mechanism. An actual experiment need not visit every displayed state.

4 Estimation and a small optional experiment

Existing rounds first. Use the implemented randomized score

$$R_t = \left[\frac{U_t}{e_t} - \frac{1 - U_t}{1 - e_t} \right] (x_{t+1} - x_t). \quad (9)$$

Use sampled activation, its logged probability, and proof share before the action. Verify forgetting-boundary timing. Correct sequential randomization and both actions having positive probability make the conditional mean score the immediate activation effect; no scalar- x Markov assumption is needed.

Within one task, target, budget and persistence condition, fit

$$R_t = \alpha_b + \beta_b \phi_t + \text{terms for current } x_t \text{ and round} + \epsilon_t. \quad (10)$$

The fixed-environment toy predicts $\alpha_b = w_b \Lambda \delta_L$ and $\beta_b = w_b \Lambda (\delta_P - \delta_L)$. Extra terms diagnose departures and composition effects. Compare fitted and raw binned responses where ϕ varies sufficiently. Avoid extrapolating an intercept from data far from $\phi = 0$. The ratio $r = 1 + \beta_b / \alpha_b$ requires the toy assumptions and a supported nonzero intercept; these coefficients alone do not identify γ and both preferences separately.

Keep eight x bins and at most two or three proof-share bands initially. Bootstrap and split prediction checks by complete initialization/randomness blocks. Do not group the causal score by realized exposure or proof acquired after activation. Measured exposure can diagnose w_b ; an initially empty peer board is not a valid fixed $B = 0$ for a growing round.

Optional isolated probe: 96 participant calls. Construct 12 paired contexts for the same task. Each pair has an incomplete active fact set and a superset that symbolically determines the gold answer. Hold previous vote, public context, answer ordering and target fixed within each pair. Select a fixed truthful controller packet prospectively that adds no new fact in either condition. This primarily tests response to reporting, rather than new proof acquisition.

For each context run:

	Silence	Packet directly visible
Incomplete evidence	2 draws	2 draws
Complete proof	2 draws	2 draws

The total is $12 \times 2 \times 2 \times 2 = 96$ calls with the same participant model, response contract and decoding settings. Direct visibility deliberately sets exposure to one; it does not reproduce board sampling. Use fresh calls without conversation carry-over. This is a modest pilot, not a precise individual-response estimate.

Estimate $d_g = \Pr(\text{target} \mid \text{packet}, g) - \Pr(\text{target} \mid \text{silence}, g)$ and the paired contrast $d_P - d_L$. Under the common-compliance toy, $d_g = \gamma \delta_g$. Thus $d_P < d_L$ supports protection; if d_L is well separated from zero, d_P / d_L estimates r . Bootstrap the 12 paired contexts. Proof additions also change prompt length and content: interpret the contrast as the effect of these specified evidence additions, not a universal effect of abstract knowledge. If eligible contexts or packets do not exist, report that limitation.

Decision. Retain the toy if it predicts held-out susceptibility and the probe supports weaker response with proof. Otherwise report the discrepancy before adding state variables. This note extends only susceptibility and the supplied report’s availability normalization; it makes no additional efficiency or thermodynamic claims.